Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/20/2019
PI Name:	Koochesfahani, Manoochehr Ph.D.		
Project Title:	ZBOT Thermometry and Velocimetry Diagnostics Development		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	02/05/2016	**End Date:**	12/31/2019
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2019 per NSSC; original end date was 2/14/2018 and had been extended to 7/14/2018 in Feb 2018 and then to 12/31/2018 in June 2018 (Ed., 3/28/19) NOTE: End date changed to 12/31/2018 per NSSC; original end date was 2/14/2018 and had been extended to 7/14/2018 in Feb 2018 (Ed., 7/31/18)		
Key Personnel Changes/Previous PI:	April 2019 report: Dr. Shahram Pouya moved to Iowa State University in 2018 and is no longer CoInvestigator on the project.		
COI Name (Institution):			
Grant/Contract No.:	NNX16AD52A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In-orbit or on-surface cryogenic storage for propellant and life support consumables is essential to all stages of NASA's Exploration Program. However, heat leakage from the surroundings results in liquid vaporization and significant mass loss and/or significant increase in the storage tank pressure. To mitigate the unwanted effects of cryogen vaporization, the Zero Boil-Off (ZBO) concept is being considered as an innovative approach for ventless means of controlling tank pressure through the application of active heat removal and forced liquid jet mixing. The Zero Boil-Off Tank (ZBOT) experiment is designed for operation on the International Space Station (ISS). Its purpose is to examine the ZBO concept and, along with computational modeling efforts, provide information on the fundamental physics of the underlying complex thermofluids environment, which is needed for engineering design optimization of the ZBO concept. The ZBOT experiment currently relies on Digital Particle Image Velocimetry (DPIV) to obtain flow field information and visualization in the test fluid, Perfluoro-n-Pentane (PnP). Difficulties have been encountered, however, in using DPIV. These difficulties include a drastic loss of particles over time and decrease of fluorescence intensity over time from the fluorescent-coated particles, along with concerns over particle sticking and agglomeration. The proposed effort is motivated by the need to obtain non-intrusive, whole-field, temperature and/or velocity data using optical diagnostics in the ZBOT microgravity experiment in order to supplement the computational modeling efforts in progress. Because of the challenges experienced by the ZBOT team in using particle-based techniques (i.e., DPIV), our focus will be on development of molecular-based approaches for measurements within PnP. These include laser induced fluorescence (LIF) methods for thermometry and molecular tagging approaches for velocimetry (MTV).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	DECEMBER 2020 The ZBOT experiments completed aboard the International Space Station (ISS) were designed to study certain issues connected to storage tank self-pressurization. The experiments also explored novel approaches to ventless tank pressure control under microgravity conditions; these approaches involved the use of active heat removal and forced liquid jet-mixing. As part of these experiments, particle imaging was used to interrogate the characteristics of fluid flow under various test conditions. Particle image data, using particle streak visualization and/or digital particle image velocimetry (DPIV), can provide important qualitative and quantitative velocity field information. The particle image sequences acquired in the ISS ZBOT experiments constituted a raw data set of about 70GB of particle image data, covering a range of experimental conditions. From the ISS data set, the research team created movie sequences of particle streak images over time. Movie series were created for all of the ZBOT experiments that were carried out on the ISS. This mode of analysis of flow field evolution is particularly effective when the particle density is too low to be optimum for DPIV processing. The constructed particle streak data set for all the ISS experiments amounted to approximately 150GB of data. The computed DPIV results for all the ZBOT data sets have been shared with the NASA team at Glenn Research Center. MAY 2019 The original objective of the project was to develop non-intrusive molecular-based temperature and/or velocity measurement methods suitable for the ZBOT microgravity experiment. This was partly motivated by the potential challenges connected to the use of particle-based techniques (i.e., DPIV). In addition, whole-field thermometry capability for the ZBOT experiment did not exist at the time. At the conclusion of this two-year effort, we have now demonstrated the feasibility of molecular tagging velocimetry (MTV) and also laser induced fluorescence (LIF) thermometry based on solubilized quantum dots (QD). The current annual report is focused on the additional work that was subsequently undertaken to process the particle image sequences obtained in the ISS ZBOT experiments in order to obtain quantitative information about the flow field. Here we consider both the particle streak mode of data interpretation and DPIV mode of data processing. The type and quality of particle image data determine which mode of analysis will lead to most useful data for interpretation and understanding of the flow field. (Ed. note December 2021: compiled from PI's technical report submitted April 2019 and December 2020.)
Bibliography Type:	Description: (Last Updated: 03/02/2026)
Abstracts for Journals and Proceedings	Pouya S, Olson D, Blanchard G, Koochesfahani M. "Progress in molecular based thermometry and velocimetry for the ZBOT experiment." Presented at the 71st Annual Meeting of APS Division of Fluid Dynamics, Atlanta, Georgia, November 18-20, 2018. Bulletin of the American Physical Society. 2018 Nov;63(13):Abstract ID: BAPS.2018.DFD.L21.8. http://meetings.aps.org/link/BAPS.2018.DFD.L21.8 , Nov-2018
Abstracts for Journals and Proceedings	Koochesfahani M, Pouya S, Olson D, Blanchard G. "Development of whole-field planar optical diagnostics for thermometry in the ZBOT experiment." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Program and abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018